



Introduction

Electrical & Electronic Circuits and Elements

Embedded Systems & Embedded Programming

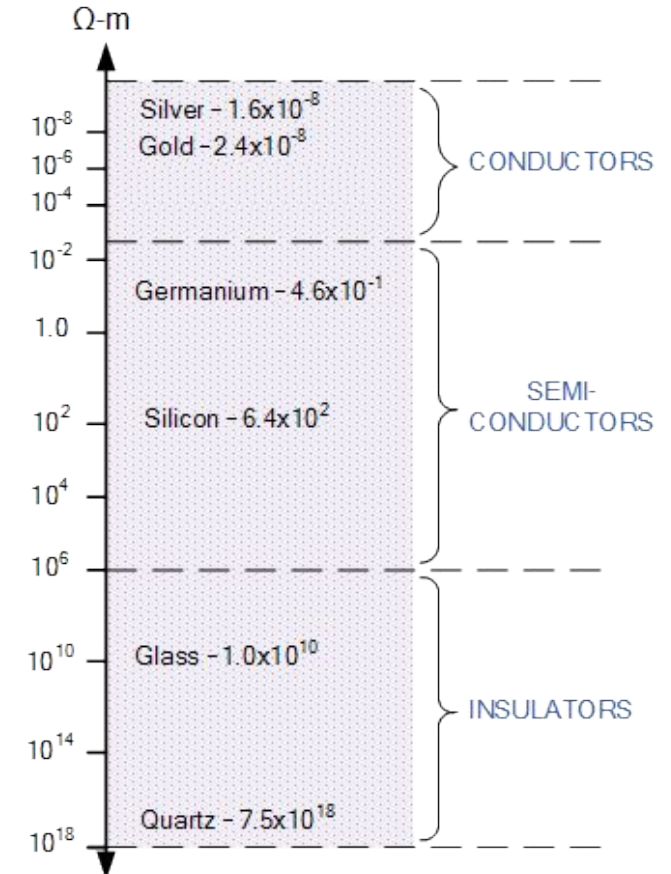
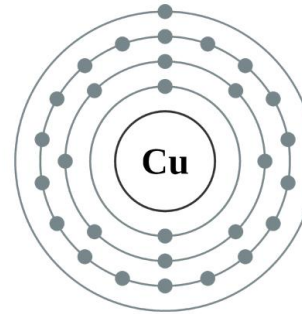
Fundamentals of Electronic Circuits (Conductors)

❖ Materials groups based on their resistivities

- Conductors
- Semiconductors
- Insulators

❖ Conductors

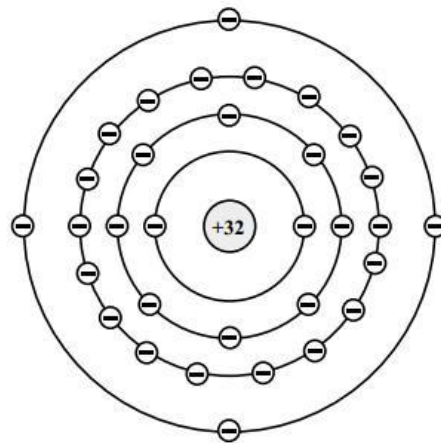
- Have very low resistivities (high conductivities)
- Electrons are the charge carriers
- Have less than 4 electrons in their valence shell
- Electrons in the valence shell can easily leave the atoms
- Current can easily flow through them because of free electrons
- By increasing temperature, their resistance increases
- Examples: Silver, Copper and Gold. All have one valence electron.



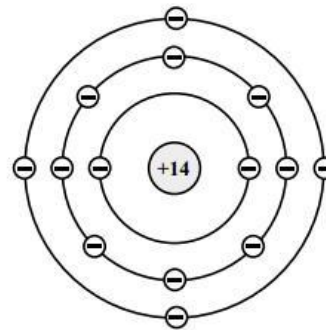
Resistivity chart of common materials

Fundamentals of Electronic Circuits (Semiconductors)

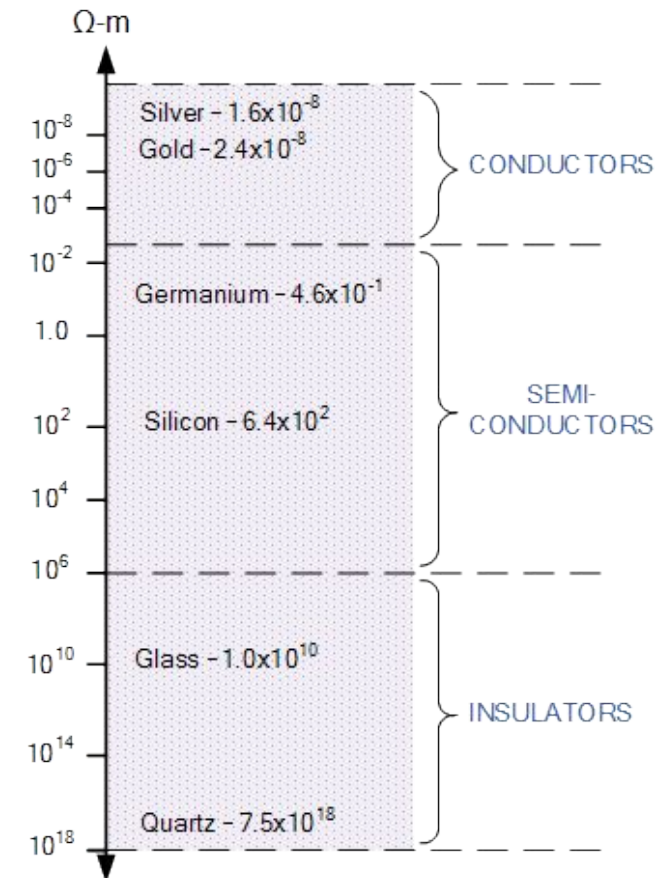
- ❖ Their electrical properties are between conductors and insulators
 - Moderate resistivity and medium conductivity
 - They are not good conductors nor good insulators
- ❖ Have 4 electrons in the valence shell
- ❖ Examples: Silicon and Germanium.



Germanium Atom

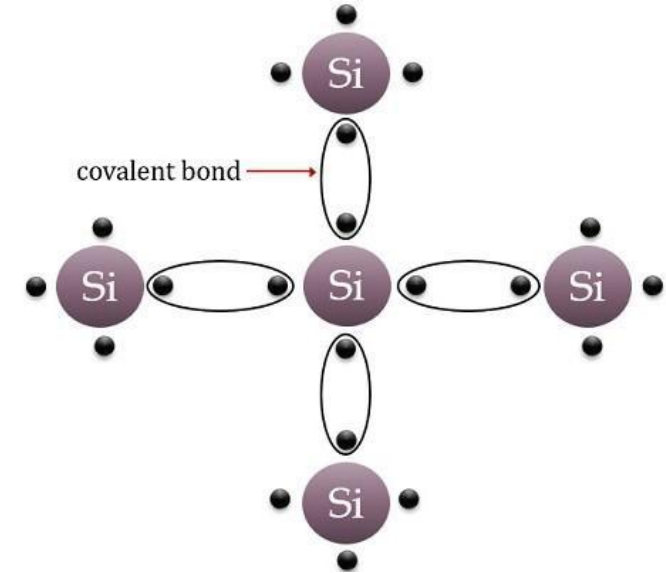


Silicon Atom



Fundamentals of Electronic Circuits (Semiconductors)

- ❖ The charge carriers are electrons and holes
- ❖ By increasing temperature, their resistance decreases
- ❖ Their atoms are closely grouped together in a crystal lattice
- ❖ Electrons in the valence shell can not easily leave the atoms
 - There are very few free electrons/holes
 - And therefore their resistance is high
- ❖ To improve and control their conductivities we can
 - Replace or add certain atoms to their pure crystal structures
 - These atoms (e.g. **Boron** or **Phosphorus**) are called **impurities**
 - The impurities produce more free electrons or holes
 - This process is called **doping**
- ❖ **Silicon** is the most used semiconductor



Crystal lattice of pure Silicon



[Intrinsic Semiconductors](#)



[Tutorial: Doping](#)

Fundamentals of Electronic Circuits (Semiconductors)

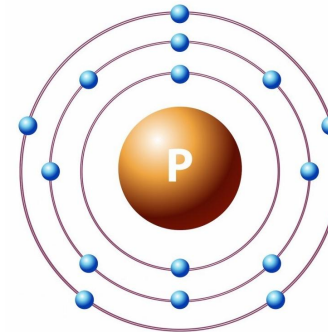
❖ **N-type** Semiconductors (e.g. doped with Phosphorus)

- Phosphorus has 5 electrons in its valence shell
- 4 of the 5 electrons make covalent bonds with its neighbouring silicon atoms
- And leaves one “free electron” to become mobile
- These impurities are known as **Donors**
- The **Donors** are positively charged

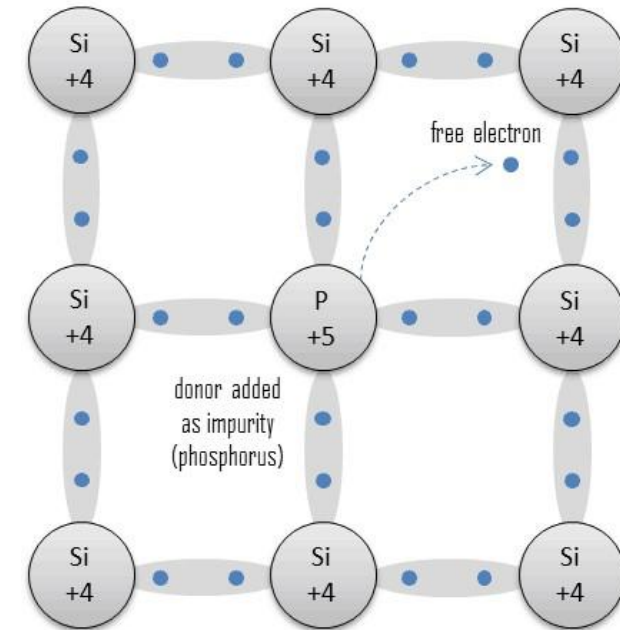


❖ **N-type** properties

- There are a large number of free electrons
- Electrons are the majority carriers
- Holes are the minority carriers
- Electrons are negatively charged
- Free electrons negatively charged



Phosphorus atom
Valence electrons: **5**
Atomic number: **15**



Fundamentals of Electronic Circuits (Semiconductors)

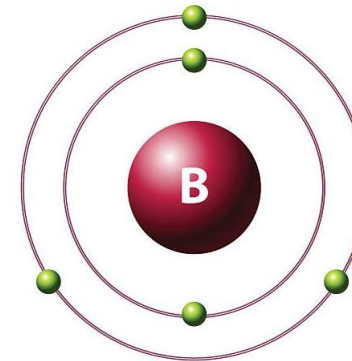
❖ **P-type** Semiconductors (e.g. doped with Boron)



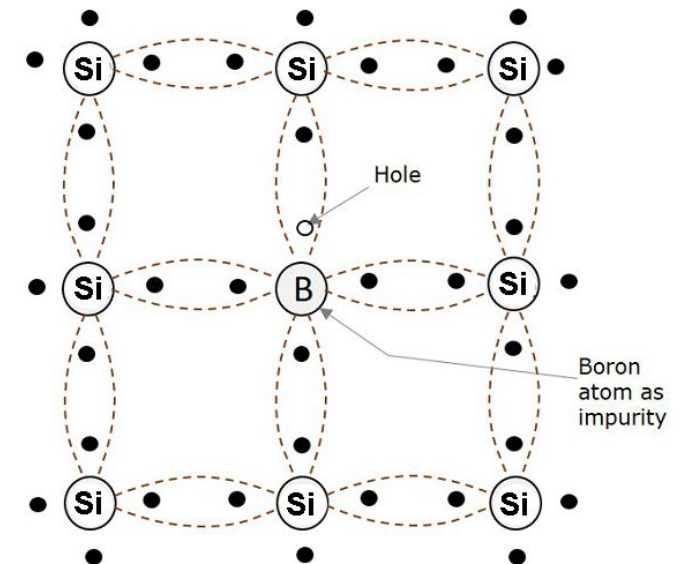
- Boron has 3 electrons in its valence shell
- Valence electrons form covalent bonds with its neighbouring silicon atoms
- Gives the semiconductor a positively charged carrier known as **hole**
- These impurities are known as **Acceptors**
- The **Acceptors** are negatively charged

❖ **P-type** properties

- There are a large number of holes
- Holes are the majority carriers
- Electrons are the minority carriers
- Holes are positively charged
- Free electrons negatively charged

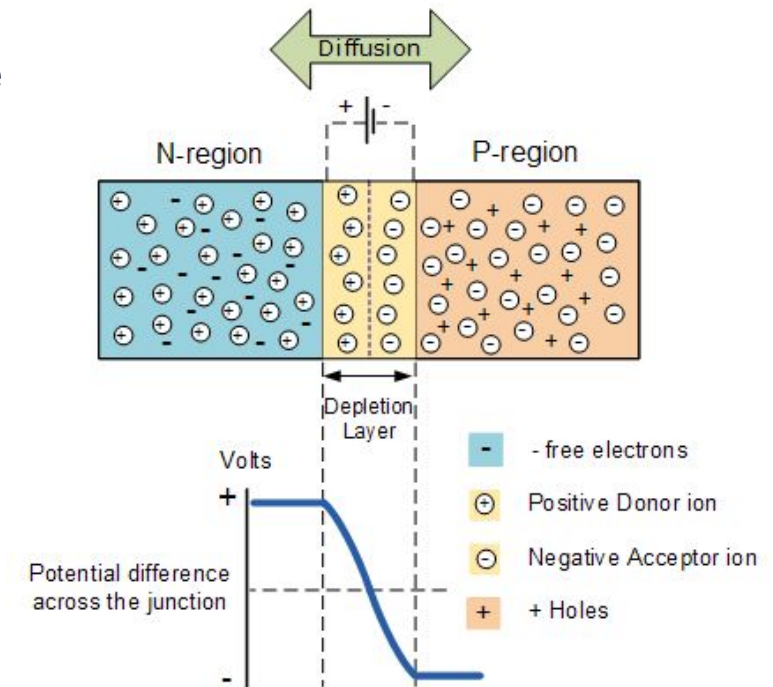


Boron atom
Valence electrons: 3
Atomic number: 5



Fundamentals of Electronic Circuits (PN Junction)

- ❖ **PN Junction** is produced by fusing (joining) a p-type and a n-type semiconductors
- ❖ Some of the **free electrons** start moving across the junction from the **N-type** to the **P-type** silicon
 - They fill some holes in the P-type material
 - They produce some negatively charged ions in the P-type
 - They leave some positively charged ions in the N-type
- ❖ The charge transfer of electrons and holes across the PN junction is known as **diffusion**
- ❖ Eventually diffusion is stopped when the electrons crossed the junction have a large enough electrical charge to repel or prevent any more charge carriers from crossing the junction



Fundamentals of Electronic Circuits (PN Junction)

❖ A **potential barrier** (electric field) zone around the junction is formed

- The donor atoms repel the holes
- The acceptor atoms repel the electrons



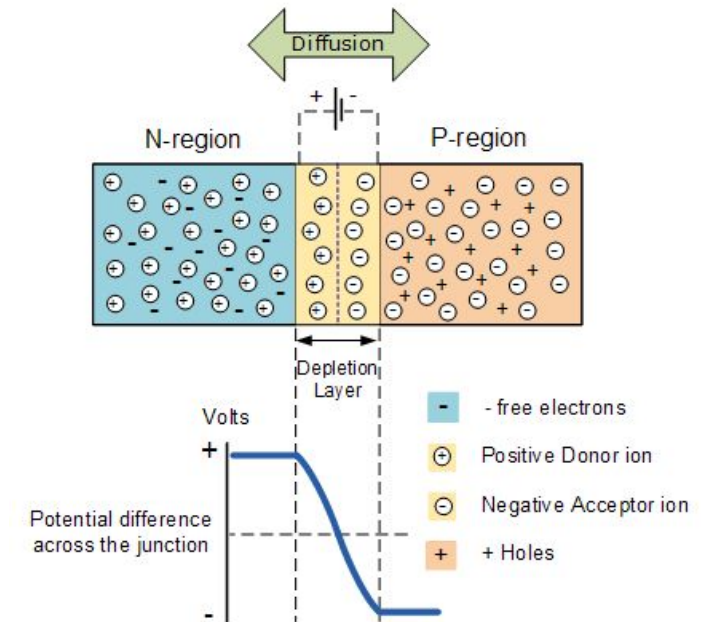
❖ A region with no free charge carriers (electron/holes) is formed around the junction

- This region is called **depletion region**
- The voltage across the depletion region at 25 °C for
 - Silicon is about 0.6 – 0.7 volts
 - Germanium is about 0.3 – 0.35 volts

❖ The **potential barrier** will always exist

- Even if the device is not connected to any external power source

❖ This device is called PN junction **Diode**



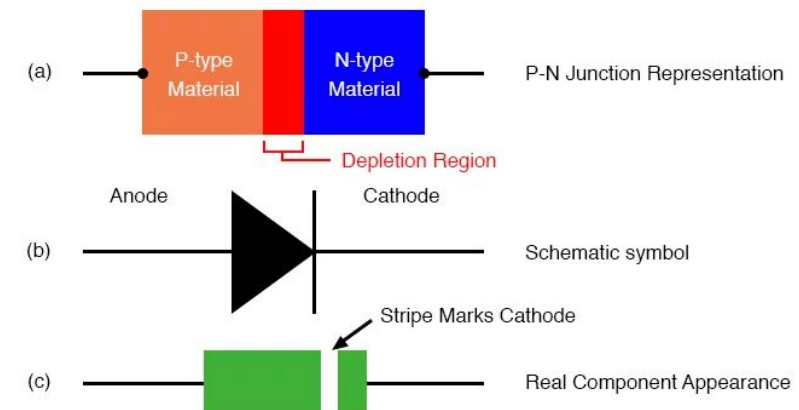
Fundamentals of Electronic Circuits (Diodes)

❖ Diode is a non-linear active element that controls the flow of current

- Current can only flow through in one direction (**forward direction**)
- Current trying to flow the **reverse direction** is blocked
- If the voltage across a diode is negative (**reverse biased**), no current can flow
 - Ideally it looks like an open circuit (OFF)
- If the positive voltage across the diode (**forward biased**) is more than the forward voltage
 - It conducts current and ideally acts like a short circuit (ON)

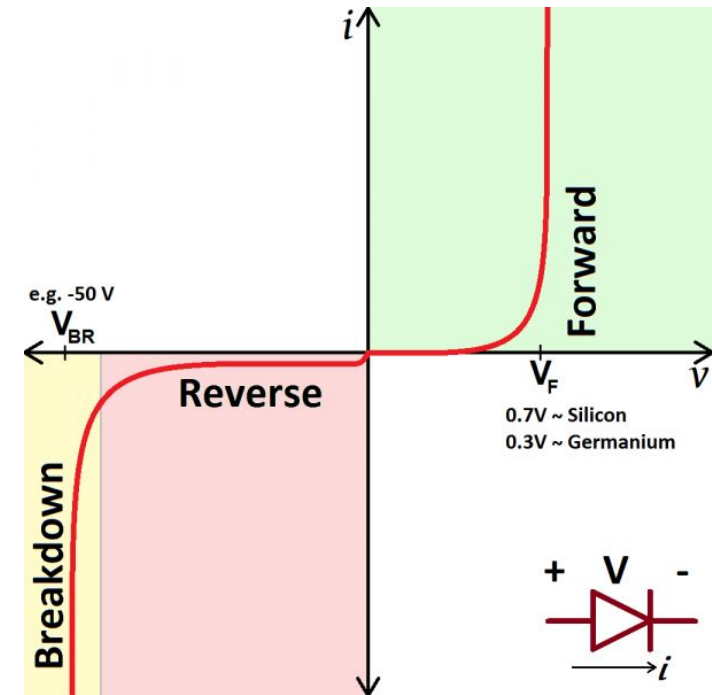
❖ Types

- Signal Diode (Normal Diode)
- Rectifier or power diode
- Light-Emitting Diode (LED)
- Schottky Diode
- Zener Diode
- Photodiode



Fundamentals of Electronic Circuits (Diodes, i-v)

- ❖ A diode operates in one of three regions:
- ❖ **Forward bias:** The applied voltage is positive
 - Current can flow through the diode. In order to have a significant current, the voltage should be greater than the forward voltage
- ❖ **Reverse bias:** The applied voltage is negative
 - In this mode current flow is (mostly) blocked
 - A very small amount of current (few nA) is able to flow
 - This current **reverse saturation current**
- ❖ **Breakdown:** The applied voltage is very large and negative
 - Lots of current will flow in the reverse direction, from cathode to anode
 - This large negative voltage is called the **breakdown voltage**
 - The breakdown voltage normally is around -50V to -100V, or more



[How does a Diode Work?](#)

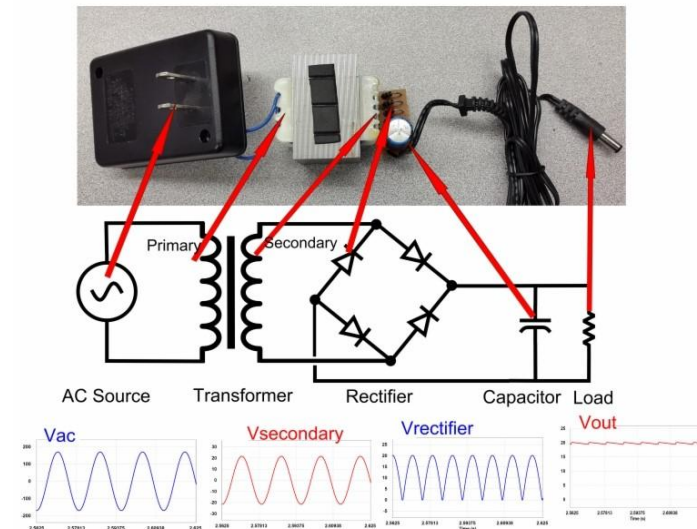
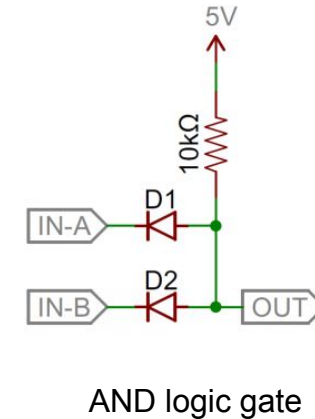
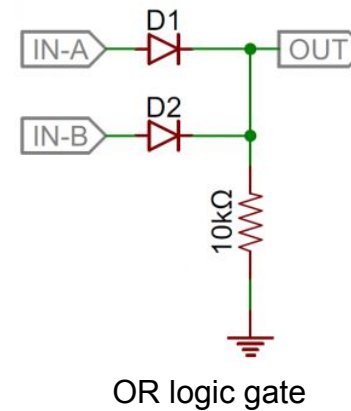
Fundamentals of Electronic Circuits (Diodes)

❖ Diode Applications

- Rectifiers
- Reverse Current Protection
- Logic Gates
- Voltage Spike Suppression (Flyback Diodes)
- Sensor (Photodiode)

❖ Useful links:

- [Sparkfun - Diodes](#)
- [How do Solar cells work?](#)
- [LED Basics](#)
- [The LED - How LEDs work?](#)
- [Diodes - What Are Diodes?](#)
- [Working Principle of Diode](#)



Fundamentals of Electronic Circuits

❖ Some useful links

- [Semiconductor Basics](#)
- [Basic Introduction, N type vs P type Semiconductor](#)
- [Semiconductor introduction](#)
- [Electrons and Holes in Semiconductors](#)
- [Extrinsic Semiconductors](#)
- [Light Emitting Diode \(LED\) Working Principle](#)
- [How diodes, LEDs and solar panels work](#)
- [How Does an LED Work?](#)
- [Effect of Temperature on Resistance](#)